Galatea: Merging Local and Global Descriptors with Optimal Transport for Visual Place Recognition on Extraterrestrial Surfaces

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***Abstract*—Visual place recognition (VPR) on the Moon has to cope with black shadows that move with the Sun, statistically self-similar crater fields and strongly non-Lambertian regolith. We present *Galatea*, a two-stage VPR system built on optimal transport (OT). Stage 1 aggregates foundation-model tokens with a semi-relaxed unbalanced OT assignment whose relaxation parameter λ recovers the NetVLAD ($\lambda = 0$) and SALAD ($\lambda = 1$) assignments exactly. We prove that its truncated Sinkhorn iterations converge linearly at rate λ, characterise the dust-bin as a logistic gate on cluster affinity, and show that after intra-normalisation the descriptor depends on the transport plan only through per-token weights. Stage 2 re-ranks candidates with a training-free partial fused Gromov–Wasserstein matching of keypoints, and a saturating score fuses this local evidence with the global similarity. We also introduce LunarSim-VPR, a physically motivated procedural benchmark in which the same query poses are rendered under seven Sun conditions. Terrestrial state-of-the-art models fall from about 66% Recall@1 at fixed illumination to 23.4% on the hardest Sun changes. Lunar-adapted OT aggregation raises overall Recall@1 from 35.8% to 59.8%, and global–local fusion adds a further 6.3 ± 0.8 points (three seeds). Consistent with the theory, balanced and unbalanced assignments perform alike. How local and global evidence are fused matters more than which local matcher is used.**

***Index Terms*—Localization, visual place recognition, space robotics, optimal transport, deep learning for visual perception.**

I. Introduction

Surface missions to the Moon need to know where they are without GNSS. Wheel and visual odometry [1] drift without bound, so a rover periodically needs an absolute fix or a loop closure. One way to get it is to recognise that the current view shows a place that is already in a reference set, either imagery from an earlier traverse or views rendered from orbital digital elevation models (DEMs) [2], [3]. This is *visual place recognition* (VPR), usually posed as image retrieval against a geo-tagged database [4], [5].

Modern VPR systems reach high recall on street-level benchmarks by aggregating deep local features into a compact global descriptor [6], [7], [8], [9], [10], optionally followed by local-feature re-ranking [11]. The lunar surface removes almost every cue these systems rely on:

- **Illumination.** There is no atmosphere to scatter light, so shadows are black and sharp. Over a lunar day the Sun's elevation and azimuth sweep through large ranges, and the same place can look entirely different at two times (Fig. 3).
- **Self-similarity.** Craters and boulders follow power-law size–frequency distributions [12], [13], so terrain is statistically repetitive at every scale and individual features are weakly discriminative [14].
- **Photometry.** Regolith is strongly non-Lambertian, with phase-dependent limb darkening and an opposition surge [15], [16], which violates the brightness-constancy assumptions built into terrestrial descriptors.
- **Data.** There is no large, pose-annotated lunar VPR dataset to train on.

This paper presents **Galatea**, a two-stage lunar VPR system built on optimal transport (OT) at both stages. Its global stage aggregates foundation-model tokens with an OT assignment, and its local stage re-ranks candidates with an OT matching that also respects geometry. Our contributions are:

1) **A semi-relaxed unbalanced OT aggregator** (Sec. IV-B). We show that one learnable scalar $\lambda \in [0, 1]$ interpolates *exactly* between the NetVLAD soft-assignment ($\lambda = 0$) and the balanced Sinkhorn assignment of SALAD [10] ($\lambda = 1$). We prove linear convergence with rate λ for the truncated iterations and characterise the dust-bin as a logistic gate on cluster affinity (Props. 1 to 3).
2) **Global–local fusion with partial fused Gromov–Wasserstein re-ranking** (Secs. IV-D and IV-E). A training-free local stage scores both the appearance and the relative layout of keypoints and is invariant to similarity transforms (Prop. 5). A saturating fusion combines it with the global similarity. We also show that the column-scale invariance of intra-normalised descriptors (Prop. 4) explains why balanced and unbalanced assignments behave alike.
3) **LunarSim-VPR** (Sec. V)[1]. This is an open, procedurally generated benchmark with a physically motivated renderer (crater size–frequency statistics, lunar-Lambert photometry with opposition surge, cast shadows from local and regional relief). Its protocol separates viewpoint change from controlled changes in solar elevation and azimuth.
4) **A controlled evaluation** (Sec. VI). We evaluate terrestrial state-of-the-art models zero-shot, compare aggregation heads trained under identical conditions, and test four re-ranking strategies, with seed variation and paired-bootstrap confidence intervals. The results show

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[1]Code and data: https://github.com/RaunakDune/galatea-lunar-vpr

that illumination-diverse lunar training and global–local fusion drive the gains, while the choice of assignment relaxation and of local matcher matters little.

II. Related Work

Global descriptors for VPR: VLAD [17] and its trainable counterpart NetVLAD [6] softly assign local features to clusters and sum their residuals. GeM pooling [7] and classification-based training at scale (CosPlace [8], EigenPlaces [9]) give strong compact descriptors. MixVPR [18] mixes feature maps with MLPs. AnyLoc [19] showed that unsupervised aggregation of DINOv2 [20] features generalises across domains, including aerial and underwater imagery. SALAD [10] recast NetVLAD's assignment as balanced entropic OT with a dust-bin on top of a partially fine-tuned DINOv2 backbone. Our aggregator generalises SALAD to unbalanced OT and contains NetVLAD's assignment as a limiting case.

Re-ranking and matching: Two-stage pipelines re-rank global candidates with local evidence, for example multi-scale patch descriptors [11] or keypoint matching with geometric verification. Learned matchers such as SuperGlue [21] and LightGlue [22] solve (or approximate) an OT problem with dust-bins over attention-refined SuperPoint [23] features. They are trained on terrestrial imagery and assume that the descriptors carry enough appearance information. Our re-ranker is training-free. It adds a Gromov–Wasserstein structural term [24], [25], in the fused form of [26] and with partial mass [27], so that matches must preserve relative keypoint geometry.

Optimal transport: Entropic regularisation and the Sinkhorn algorithm [28], [29] made OT differentiable and fast. Unbalanced OT replaces hard marginal constraints with divergence penalties and keeps Sinkhorn-like scaling algorithms [30]. The Earth Mover's Distance [31] brought OT to image retrieval.

Planetary localisation: Rovers have relied on visual odometry [1]. Landing systems use terrain-relative navigation against orbital maps [2]. Recent work renders synthetic optical images from DTMs for lunar navigation [3] and retrieves lunar crater images with visual and depth features [32]. Surface-level lunar place recognition under controlled illumination change has received little systematic study. LunarSim-VPR is meant as a reproducible test bed for it.

III. Problem Formulation

Let $\mathcal{D} = \{(\mathbf{I}_r^{\mathrm{db}}, \mathbf{x}_r, \psi_r)\}_{r=1}^{N}$ be a database of reference images with known planar position $\mathbf{x}_r \in \mathbb{R}^2$ and heading ψ_r, and let $\mathbf{I}^q$ be a query image taken at an unknown pose $(\mathbf{x}^q, \psi^q)$, generally under a different illumination. A database image is a *true match* of the query if $\|\mathbf{x}_r - \mathbf{x}^q\|_2 < \rho$ and $|\psi_r - \psi^q|_\circ < \vartheta$, where $|\cdot|_\circ$ is the wrapped angular difference (we use $\rho = 10\,\mathrm{m}$, $\vartheta = 60^\circ$). VPR is posed as retrieval [4], [5]: an encoder Φ maps every image to a unit-norm global descriptor and the database is ranked by cosine similarity $s_g(q, r) = \langle \Phi(\mathbf{I}^q), \Phi(\mathbf{I}_r^{\mathrm{db}}) \rangle$. A two-stage system additionally re-scores the top-K candidates with a local score $s_\ell(q, r)$ computed from keypoint correspondences. Performance is measured by Recall@k, the fraction of queries with at least one true match among the top-k results:

$$\mathrm{R@}k = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \mathbf{1}\big[\exists\, r \in \mathrm{top}_k(q) :\ r \text{ is a true match of } q\big]. \tag{1}$$

IV. Method

Fig. 1 gives an overview. Stage 1 aggregates the patch tokens of a vision transformer into a global descriptor by solving a *semi-relaxed unbalanced* optimal-transport (UOT) problem between tokens and a set of learned clusters plus a dust-bin (Sec. IV-B). Stage 2 re-ranks the K best candidates with a *partial fused Gromov–Wasserstein* (FGW) matching of keypoints, which scores both what the keypoints look like and how they are arranged (Sec. IV-D). The final ranking fuses the global and local scores (Sec. IV-E).

A. Tokens, scores and the assignment problem

A DINOv2 ViT-B/14 backbone [20] produces n patch tokens $\mathbf{t}_1, \ldots, \mathbf{t}_n \in \mathbb{R}^d$ and a class token $\mathbf{t}_{n+1}$ ($d = 768$; $n = 300$ for a 210×280 input). Following [10], two small MLPs map each token to an assignment score vector $\mathbf{s}_i = (s_{i1}, \ldots, s_{im}) \in \mathbb{R}^m$ over m clusters and to a reduced feature $\mathbf{f}_i \in \mathbb{R}^\ell$:

$$\begin{aligned} \mathbf{s}_i &= \mathbf{W}_{s_2}\, \sigma(\mathbf{W}_{s_1} \mathbf{t}_i + \mathbf{b}_{s_1}) + \mathbf{b}_{s_2}, \\ \mathbf{f}_i &= \mathbf{W}_{f_2}\, \sigma(\mathbf{W}_{f_1} \mathbf{t}_i + \mathbf{b}_{f_1}) + \mathbf{b}_{f_2}. \end{aligned} \tag{2}$$

Unlike NetVLAD [6], the score layer is randomly initialised (no k-means centroids), so no geometric prior on the clusters is imposed. A learnable scalar z scores every token against an extra *dust-bin* column, giving the augmented score matrix $\bar{\mathbf{S}} = [\mathbf{S} \,|\, z\mathbf{1}_n] \in \mathbb{R}^{n \times (m+1)}$. Each token carries unit mass, $\mathbf{a} = \mathbf{1}_n$, and the clusters and dust-bin have nominal capacities $\mathbf{b} = (\mathbf{1}_m^\top, n - m)^\top$ (this requires $n > m$). We write $H(\mathbf{P}) = -\sum_{ij} P_{ij}(\log P_{ij} - 1)$ for the entropy and $\mathrm{KL}(\mathbf{x} \| \mathbf{y}) = \sum_j x_j \log(x_j / y_j) - x_j + y_j$.

a) Two known special cases: NetVLAD distributes each token's mass independently over the clusters with a softmax, $P_{ij} = \mathrm{softmax}_j(s_{ij}/\varepsilon)$. This constrains only the rows of $\mathbf{P}$ (feature→cluster) and ignores how much mass each cluster receives (cluster→feature). SALAD [10] constrains both marginals by solving the balanced entropic OT problem [28]

$$\mathbf{P}^{\mathrm{bal}} = \underset{\mathbf{P} \geq 0,\ \mathbf{P}\mathbf{1} = \mathbf{a},\ \mathbf{P}^\top \mathbf{1} = \mathbf{b}}{\arg\max} \langle \bar{\mathbf{S}}, \mathbf{P} \rangle + \varepsilon H(\mathbf{P}). \tag{3}$$

At convergence, with $n = 300$ and $m = 64$, (3) puts exactly one unit of mass into every cluster and $n - m = 236$ units (79% of all mass) into the dust-bin, regardless of image content. A priori this is a questionable prior on the lunar surface, where the fraction of uninformative tokens (black sky, featureless regolith, saturated slopes) varies widely with the horizon position and the Sun elevation. In practice SALAD truncates Sinkhorn after $L = 3$ iterations, which relaxes the constraint somewhat (Sec. VI-C). The formulation below makes the relaxation explicit and controllable.

B. Galatea: semi-relaxed unbalanced OT assignment

We keep the token constraint hard (every token distributes exactly its unit mass) but replace the cluster/dust-bin constraint with a KL penalty of strength $\tau > 0$ [30]:

$$\mathbf{P}^{\tau} = \underset{\mathbf{P}\geq 0,\ \mathbf{P}\mathbf{1}_{m+1}=\mathbf{a}}{\arg\max} \ \langle \bar{\mathbf{S}}, \mathbf{P}\rangle + \varepsilon H(\mathbf{P}) - \tau\,\mathrm{KL}\big(\mathbf{P}^{\top}\mathbf{1}_n \,\big\|\, \mathbf{b}\big). \tag{4}$$

Define $\lambda = \tau/(\tau+\varepsilon) \in (0,1)$ and the log-sum-exp $\mathrm{LSE}_j(x_j) = \log \sum_j e^{x_j}$. Proofs of all propositions are given in the supplementary material.

Proposition 1 (Structure, limits)**.** *Problem* (4) *has a unique solution of the form* $P^{\tau}_{ij} = \exp\big((\bar{S}_{ij} + f_i + g_j)/\varepsilon\big)$*, where the potentials* $(\mathbf{f}, \mathbf{g})$ *satisfy*

$$f_i = \varepsilon \log a_i - \varepsilon\,\mathrm{LSE}_j\big((\bar{S}_{ij} + g_j)/\varepsilon\big), \tag{5}$$

$$g_j = \lambda\big[\varepsilon \log b_j - \varepsilon\,\mathrm{LSE}_i\big((\bar{S}_{ij} + f_i)/\varepsilon\big)\big]. \tag{6}$$

(i) As $\tau \to 0$ *(*$\lambda \to 0$*),* $\mathbf{g} \to \mathbf{0}$ *and* $P^{\tau}_{ij} \to a_i\,\mathrm{softmax}_j(\bar{S}_{ij}/\varepsilon)$*: the NetVLAD soft-assignment, augmented with a dust-bin. (ii) As* $\tau \to \infty$ *(*$\lambda \to 1$*),* (5)–(6) *become the balanced Sinkhorn fixed-point equations and* $\mathbf{P}^{\tau} \to \mathbf{P}^{\mathrm{bal}}$ *of* (3)*.*

Hence a single scalar λ interpolates continuously between NetVLAD-style and SALAD-style assignment. Galatea *learns* $\lambda = \mathrm{sigmoid}(\theta)$ jointly with the rest of the head. Alternating (5) and (6) gives Alg. 1, which is SALAD's Sinkhorn loop with the cluster-side update multiplied by λ; it costs $O(Lnm)$.

Proposition 2 (Linear convergence)**.** *Let* Φ *map* $\mathbf{g}$ *to the value of* (6) *after substituting* $\mathbf{f}(\mathbf{g})$ *from* (5)*. Then* $\|\Phi(\mathbf{g}) - \Phi(\mathbf{g}')\|_\infty \leq \lambda\|\mathbf{g} - \mathbf{g}'\|_\infty$*. For* $\lambda < 1$ *the iterates of Alg. 1 therefore converge linearly,* $\|\mathbf{g}^{(L)} - \mathbf{g}^{\star}\|_\infty \leq \lambda^L \|\mathbf{g}^{(0)} - \mathbf{g}^{\star}\|_\infty$*, at a rate independent of* ε*,* n *and* m*.*

For the balanced case ($\lambda = 1$) the same map is only non-expansive in $\|\cdot\|_\infty$. Its convergence is usually proved in Hilbert's projective metric, with a rate that depends on ε and on the dynamic range of $\bar{\mathbf{S}}$ [29]. Prop. 2 instead gives an explicit, data-independent rate for every $\lambda < 1$. Numerically the bound is tight: on random score matrices the measured error ratio between successive iterations approaches λ. With learned values of λ (Sec. VI), a small fixed L is therefore justified.

Proposition 3 (Dust-bin gating)**.** *For every* $\lambda \in [0,1]$*, after any token-side update* (5) *(in particular at the output of Alg. 1), the fraction of token* i*'s mass sent to the dust-bin is*

$$P_{i,m+1} = \mathrm{sigmoid}\Big(\tfrac{1}{\varepsilon}\big[z + g_{m+1} - \varepsilon\,\mathrm{LSE}_{j\leq m}\big((s_{ij} + g_j)/\varepsilon\big)\big]\Big), \tag{7}$$

which, for fixed potentials $\mathbf{g}$*, is strictly decreasing in every cluster score* s_{ij} *and increasing in* z*.*

The dust-bin therefore acts as a learned soft gate on the token's *soft-max cluster affinity*. Tokens that resemble no cluster, such as sky, smooth regolith or noise, are suppressed before aggregation. In the relaxed problem, the potential g_{m+1}, and with it the total discarded mass, adapts to the image instead of being fixed at $n - m$.

Algorithm 1 λ-scaled log-domain Sinkhorn (Galatea assignment)

Require: scores $\bar{\mathbf{S}} \in \mathbb{R}^{n\times(m+1)}$, marginals $\mathbf{a}, \mathbf{b}$, $\lambda \in [0,1]$, iterations L

$\mathbf{f} \leftarrow \mathbf{0}_n$, $\mathbf{g} \leftarrow \mathbf{0}_{m+1}$ ▷ ε absorbed in $\bar{\mathbf{S}}$

for $\ell = 1, \ldots, L$ **do**

  $g_j \leftarrow \lambda\,[\log b_j - \mathrm{LSE}_i(\bar{S}_{ij} + f_i)]$ ▷ clusters + dust-bin (relaxed)

  $f_i \leftarrow \log a_i - \mathrm{LSE}_j(\bar{S}_{ij} + g_j)$ ▷ tokens (hard)

return $\mathbf{P} = \exp(\bar{\mathbf{S}} + \mathbf{f}\mathbf{1}^{\top} + \mathbf{1}\mathbf{g}^{\top})$ with the dust-bin column removed

The next result shows which part of the transport plan the descriptor actually sees. It explains the empirical behaviour reported in Sec. VI-E.

Proposition 4 (Column-scale invariance)**.** *Let the descriptor be built from* $\mathbf{P}$ *by* (9) *followed by per-cluster (intra-)* ℓ_2*-normalisation. Then* Φ *is invariant to* $\mathbf{P} \mapsto \mathbf{P}\,\mathrm{diag}(\mathbf{c})$ *for any* $\mathbf{c} > 0$*. In particular, for any plan of the form of Prop. 1,* Φ *depends on the cluster potentials* $\mathbf{g}_{1:m}$ only through *the token weights* $w_i = e^{f_i/\varepsilon} = a_i / \sum_{j\leq m+1} e^{(\bar{S}_{ij}+g_j)/\varepsilon}$*:*

$$\frac{\mathbf{V}_{j,:}}{\|\mathbf{V}_{j,:}\|} = \frac{\sum_i w_i e^{s_{ij}/\varepsilon}\mathbf{f}_i}{\big\|\sum_i w_i e^{s_{ij}/\varepsilon}\mathbf{f}_i\big\|}. \tag{8}$$

Balanced and unbalanced assignments therefore differ only in how they *weight tokens*, through w_i. How much total mass each cluster receives is removed by intra-normalisation. Equation (8) also shows that NetVLAD with a dust-bin ($\lambda = 0$) is SALAD-style aggregation with the simplest weights, $w_i = 1/(\sum_j e^{s_{ij}/\varepsilon} + e^{z/\varepsilon})$.

C. Aggregation and training

With $\mathbf{P} \in \mathbb{R}^{n\times m}$ (the dust-bin column dropped), the aggregated descriptor is the matrix $\mathbf{V} \in \mathbb{R}^{m\times\ell}$ with entries

$$V_{j,k} = \sum_{i=1}^{n} P_{i,j}\, f_{i,k}, \qquad j = 1,\ldots,m, \quad k = 1,\ldots,\ell. \tag{9}$$

As in [10], centroids are not subtracted. A scene descriptor $\mathbf{g}_{\mathrm{sc}} = \mathbf{W}_{g_2}\sigma(\mathbf{W}_{g_1}\mathbf{t}_{n+1} + \mathbf{b}_{g_1}) + \mathbf{b}_{g_2}$ summarises the class token. Each row of $\mathbf{V}$ is ℓ_2-normalised (intra-normalisation), $\mathbf{g}_{\mathrm{sc}}$ is ℓ_2-normalised, and the concatenation $[\mathbf{g}_{\mathrm{sc}}; \mathrm{vec}(\mathbf{V})]$ is ℓ_2-normalised again to give $\Phi(\mathbf{I}) \in \mathbb{R}^{256+m\ell}$ (8448-D for $m = 64, \ell = 128$). We train with the multi-similarity loss and miner [33] on batches of P places $\times$ 4 views. Images of *different* places closer than 15 m are masked out of the negative set, because adjacent places along a traverse overlap visually.

D. Stage 2: partial fused Gromov–Wasserstein re-ranking

Repetitive crater and boulder fields make individual local descriptors ambiguous: two places may share many similar-looking craters. What distinguishes them is *where* those craters lie relative to one another [14]. Let $\{(\mathbf{u}_i, \mathbf{d}_i)\}_{i=1}^{p}$ and $\{(\mathbf{v}_k, \mathbf{e}_k)\}_{k=1}^{p'}$ be SuperPoint keypoints [23] (pixel positions

and unit descriptors) of the query and a candidate. We define the appearance cost $M_{ik} = 1 - \langle \mathbf{d}_i, \mathbf{e}_k \rangle$ and scale-normalised intra-image distance matrices $C^q_{ii'} = \|\mathbf{u}_i - \mathbf{u}_{i'}\|/\mu_q$ and $C^c_{kk'} = \|\mathbf{v}_k - \mathbf{v}_{k'}\|/\mu_c$, where μ is the mean pairwise distance in each image. The partial FGW problem [26], [27] trades appearance against preservation of the keypoint geometry:

$$\min_{\mathbf{T} \in \bar{\mathcal{U}}} (1-\alpha)\langle \mathbf{M}, \mathbf{T} \rangle + \alpha \sum_{i,i',k,k'} \left(C^q_{ii'} - C^c_{kk'}\right)^2 T_{ik} T_{i'k'} - \varepsilon_\ell H(\bar{\mathbf{T}}), \quad (10)$$

where $\bar{\mathbf{T}} \in \mathbb{R}^{(p+1)\times(p'+1)}$ augments $\mathbf{T}$ with dust-bins that absorb non-co-visible keypoints [21], and $\bar{\mathcal{U}}$ fixes unit mass per keypoint, with p' and p units for the bins. We minimise (10) by entropic mirror descent [25]. Each step linearises the quadratic term at the current transported part $\mathbf{T}^{(r)}$ (normalised to unit mass, with marginals $\boldsymbol{\pi}, \boldsymbol{\pi}'$):

$$\mathbf{G}^{(r)} = (\mathbf{C}^q)^{\odot 2} \boldsymbol{\pi} \mathbf{1}^\top + \mathbf{1} \boldsymbol{\pi}'^\top (\mathbf{C}^c)^{\odot 2} - 2\, \mathbf{C}^q \mathbf{T}^{(r)} \mathbf{C}^{c\top}. \quad (11)$$

(this is the gradient of the quadratic term up to a factor 2, absorbed into α). It then solves an entropic OT problem with dust-bins on the fused cost $(1-\alpha)\mathbf{M} + \alpha \mathbf{G}^{(r)}$ with the log-domain Sinkhorn algorithm. The initial plan uses $\mathbf{M}$ alone. Correspondences are mutual arg-maxima of $\mathbf{T}$ with confidence above 0.2. They are verified with a MAGSAC++ [34] fundamental-matrix fit, and the local score s_ℓ is the number of inliers.

Proposition 5 (Invariances of the structural term)**.** *The objective* (10)*, and hence s_ℓ before geometric verification, is invariant to any similarity transformation (rotation, translation, isotropic scaling) applied to the keypoint positions of either image, and to any permutation of the keypoints.*

Unlike homography- or epipolar-based scoring, the structural term thus needs no camera model. It rewards candidates in which matched keypoints keep their *relative* layout, which is exactly the cue that separates two crater fields with similar statistics.

E. Fusing global and local evidence

The final score of candidate r for query q is

$$s(q,r) = s_g(q,r) + \beta \frac{s_\ell(q,r)}{s_\ell(q,r) + \kappa}, \quad (12)$$

where the saturating map keeps one very textured candidate from dominating. β and κ are selected on the validation terrain. With $\beta \to \infty$ the method reduces to pure local re-ranking, and with $\beta = 0$ to the global ranking.

F. Complexity

Alg. 1 costs $O(Lnm)$ (about $6 \cdot 10^4$ operations for $L = 3$), which is negligible next to the ViT forward pass. Each FGW step costs $O(p^2 p' + p p'^2)$ for (11) plus $O(p\,p')$ per Sinkhorn iteration. Re-ranking K candidates with R outer steps therefore costs $O(KR\,p^3)$ for $p \approx p'$. Measured run times are reported in Sec. VI-F.

V. The LunarSim-VPR Benchmark

No public lunar surface dataset combines dense pose ground truth with systematic illumination coverage. We therefore built a lightweight, reproducible renderer that targets the three properties that make lunar VPR hard: self-similar relief, hard shadows and non-Lambertian photometry. It does not replace real data (Sec. VI-G), but it lets us vary one factor at a time with exact ground truth.

Terrain: Each scene combines a *regional* DEM (16 m/px, 16.4 km) and a *local* DEM (0.3 m/px, 614 m) blended into its centre. Both start from fractional-Brownian relief (power spectrum $\propto k^{-\beta}$, $\beta \in [2.2, 3.2]$). Craters are drawn from an equilibrium-like cumulative size–frequency distribution $N(>D) = cD^{-2}$ [13] ($D \in [60, 2500]$ m regionally and $[1.5, 120]$ m locally) and stamped with a bowl–rim–ejecta profile whose depth-to-diameter ratio (0.04–0.2) and rim height depend on a random degradation state. Fresh craters get bright albedo halos and ejecta boulders. Four to seven distant massifs (200–700 m high, 4–7.5 km away) give a horizon silhouette, as at the Apollo 17 site (Fig. 2).

Photometry and shadows: Radiance follows the lunar-Lambert function of McEwen [15], $I \propto A\,[2L(g)\cos i/(\cos i + \cos e) + (1 - L(g))\cos i]$, with phase-dependent $L(g)$. It is multiplied by a Hapke-style shadow-hiding opposition term $1 + B_0/(1 + \tan(g/2)/h)$ [16], where i, e and g are the incidence, emission and phase angles. Cast shadows are computed from both DEMs (so massifs shade the valley at low Sun), averaged over the 0.53° solar disk to produce penumbrae. A weak ambient term models earthshine and inter-reflection, and a footprint-faded micro-relief bump map adds sub-grid regolith texture.

Camera: A rover-mounted pinhole camera (70° HFOV, 320×240, 1.35–1.7 m high) is rendered by column-wise height-field ray casting, accounting for lunar curvature. The renderer applies centre-weighted auto-exposure, a soft-shoulder tone curve, gamma, shot and read noise, and 8-bit quantisation. One image takes about 0.15 s on a CPU once the shadow maps for a Sun direction have been computed.

Protocol: Each scene contains a smooth random traverse with places every 6 m. The database is rendered at one Sun direction (azimuth 135°, elevation 30°). Queries are taken at jittered poses (up to 3 m offset, $\mathcal{N}(0, 10^\circ)$ heading noise, random camera height). The *same* query poses are rendered under seven Sun conditions (Table I), so differences between conditions are paired and caused by illumination alone. The test set has three unseen scenes (3×150 database images and 3×1050 queries), the validation set one scene (100 places, 700 queries), and the training set five further scenes with 240 places $\times$ 4 views. Training views use Sun directions drawn from a per-scene pool of 16 random directions (elevation 4°–75°, any azimuth), which mimics an illumination-sweep dataset.

VI. Experiments

A. Setup

Compared methods: Zero-shot terrestrial models use their official weights: CosPlace-R50 [8] (512-D), EigenPlaces-R50 [9] (2048-D) and DINOv2-SALAD [10] (8448-D, trained

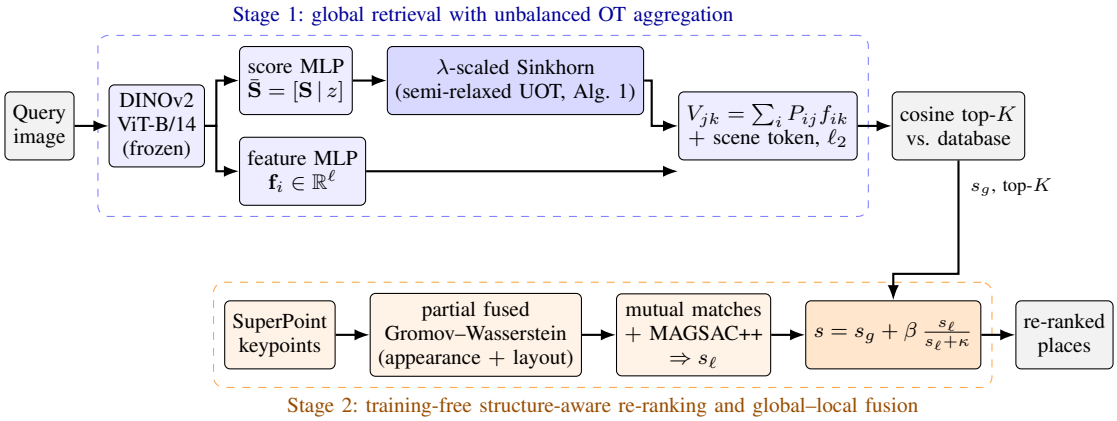


Fig. 1. Galatea. Stage 1 assigns ViT tokens to m clusters and a dust-bin by semi-relaxed unbalanced OT with a learned relaxation λ (λ=0: NetVLAD assignment, λ=1: SALAD), then aggregates them into a global descriptor. Stage 2 re-scores the top-K candidates with a partial fused Gromov–Wasserstein matching of keypoints and fuses local and global scores.

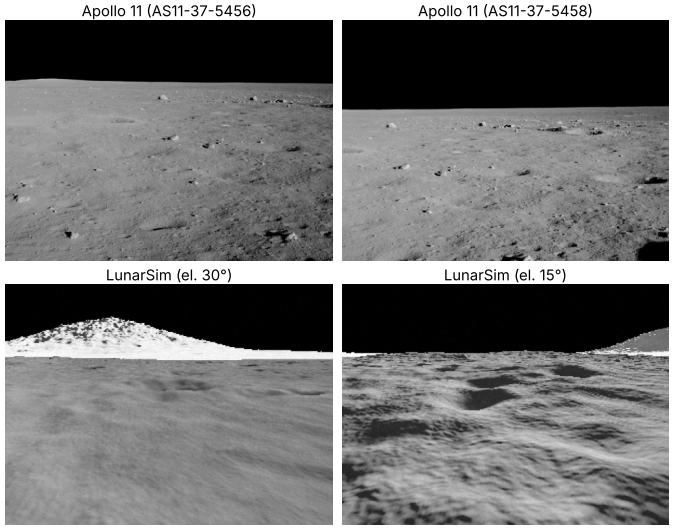


Fig. 2. Real Apollo 11 surface photographs [35] (top) and LunarSim renders (bottom). The renderer reproduces the black sky, the horizon structure and the shading statistics of low-Sun regolith. It under-resolves near-field rocks smaller than the 0.3 m DEM spacing, a limitation discussed in Sec. VI-G.

TABLE I
QUERY ILLUMINATION CONDITIONS. THE DATABASE IS RENDERED AT AZIMUTH 135°, ELEVATION 30°.

ID	Name	Az.	El.	Change w.r.t. database
C0	same	135	30	viewpoint only
C1	el15	135	15	lower Sun
C2	el6	135	6	grazing Sun, long shadows
C3	el60	135	60	high Sun, low contrast
C4	az+90	225	30	shadows rotate 90°
C5	az+180	315	30	shadows flip
C6	az+180, el8	315	8	flip + grazing (hardest)

on GSV-Cities [36]). *Lunar-trained heads* all share the frozen SALAD-fine-tuned DINOv2 ViT-B/14 backbone, the training data, the loss and the schedule, and differ only in the aggregation: GeM [7] (1280-D), NetVLAD [6] (k-means-initialised, 64×128 + scene token, 8448-D), SALAD (balanced OT, $\lambda = 1$) and Galatea (semi-relaxed UOT with learned λ). The last three have identical descriptor size and almost identical parameter counts. For stage 2 we compare four local scorers on the same SuperPoint keypoints (top 384 by score) and top-$K = 10$ candidates: mutual nearest neighbours with cosine > 0.7 (MNN), training-free entropic OT with dust-bins (UOT, appearance only), the learned LightGlue matcher [22], and our partial FGW. Each is used with and without MAGSAC++ verification.

Implementation: Images are resized to 210×280 for the ViT ($n = 300$ tokens). Heads are trained with AdamW (learning rate $4 \cdot 10^{-4}$, one-cycle schedule, weight decay 10^{-4}) for 500 steps of 24 places $\times$ 4 views, using the multi-similarity loss ($\alpha = 1$, $\beta = 50$, miner margin 0.1). Sinkhorn uses $L = 3$ iterations, $m = 64$ clusters and $\ell = 128$. The four main heads and the $\lambda = 0$ ablation are run with three seeds (mean $\pm$ standard deviation); the other ablations use one seed. For stage 2, the structural weight $\alpha \in \{0, 0.3, 0.6\}$ (chosen: 0.3) and the fusion parameters β, κ are selected on the validation scene only. $\varepsilon_\ell = 0.05$ and the dust-bin cost threshold 0.25 (cosine 0.75) are fixed a priori, and MAGSAC++ uses a 1.5 px threshold. All experiments run on a 2-core CPU without a GPU. Code, the renderer, the rendered benchmark and the trained heads are available at https://github.com/RaunakDune/galatea-lunar-vpr.

B. Terrestrial models under lunar illumination change

Table II (top) shows that all three terrestrial models perform reasonably when only the viewpoint changes: at C0, R@1 is 65–67%. They degrade sharply as soon as the Sun moves. DINOv2-SALAD, the strongest of them, falls from 66% at C0 to 21% under grazing illumination (C2) and to 25% when the shadows flip (C5). The CNN-based CosPlace and EigenPlaces fare worse, with C5 values of 10% and 16%. Even a foundation-model backbone pre-trained on 142M images therefore does not provide illumination-invariant place descriptors on regolith. Local re-ranking does not close the gap either. Fusing MNN+MAGSAC++ matches into the top-10 zero-shot SALAD candidates (selected on validation as

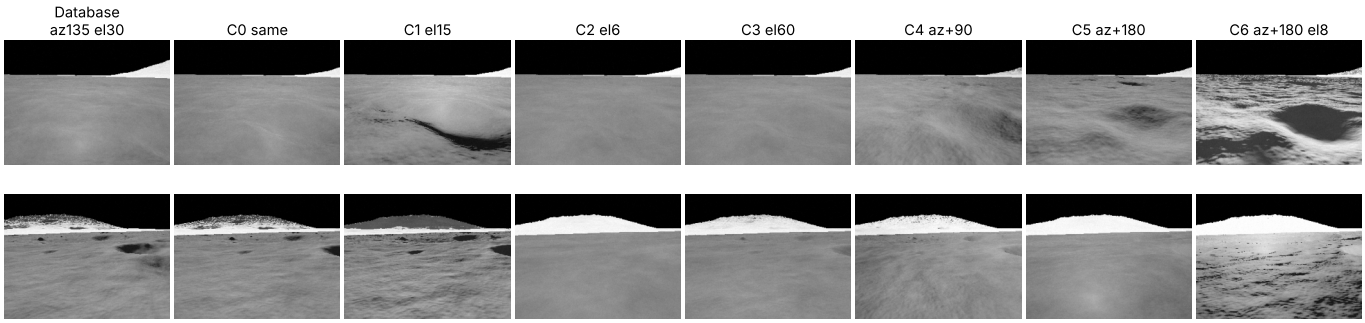


Fig. 3. LunarSim-VPR: two test places (rows). Left: database image (Sun azimuth 135°, elevation 30°). Right: queries at a jittered pose under the seven Sun conditions of Table I. Rendered with lunar-Lambert photometry, opposition surge and cast shadows from local and regional relief. In the top row at C2, the camera lies in the shadow of a distant massif and is lit only by the weak ambient term.

TABLE II
RECALL (%) ON THE LUNARSIM-VPR TEST SET (3 UNSEEN SCENES, 3150 QUERIES). LUNAR-TRAINED HEADS SHARE THE FROZEN DINOV2 BACKBONE AND TRAINING PROTOCOL; MEAN±STD OVER 3 SEEDS. PER-CONDITION COLUMNS GIVE R@1 (CONDITIONS IN TABLE I).

	Method	Dim.	R@1	R@5	R@10	C0	C1	C2	C3	C4	C5	C6
zero-shot	CosPlace-R50 [8]	512	26.8	50.3	60.8	66.9	35.8	17.6	24.9	17.6	10.4	14.4
	EigenPlaces-R50 [9]	2048	28.1	53.0	64.1	64.9	34.2	17.1	25.3	22.2	15.6	17.1
	DINOv2-SALAD [10]	8448	35.8	65.4	77.7	66.4	46.7	21.3	31.6	35.8	24.9	24.0
lunar-trained	GeM	1280	31.5±0.4	64.7±0.3	79.4±0.2	55.0	34.8	19.8	35.3	30.1	25.1	20.3
	NetVLAD	8448	58.2±0.7	86.6±0.4	94.6±0.2	79.0	67.5	46.1	65.2	56.0	48.3	45.6
	SALAD (λ=1)	8448	59.4±0.7	**87.3±0.7**	**94.8±0.2**	77.6	68.4	**46.4**	65.4	58.6	52.7	**46.7**
	Galatea, learned λ	8448	**59.8±0.8**	**87.3±0.5**	94.6±0.2	**79.3**	**69.0**	44.4	**67.3**	**60.4**	**54.7**	43.7

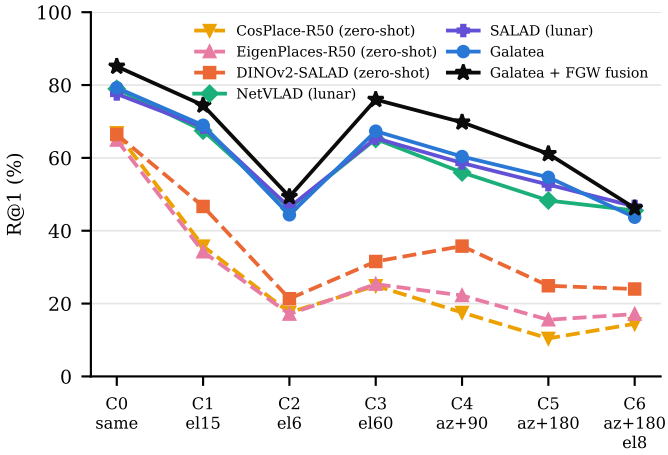


Fig. 4. R@1 per illumination condition on the test set. Zero-shot terrestrial models (dashed) collapse once the Sun moves. Lunar-trained OT aggregation (Galatea) and global–local fusion (Galatea + FGW) recover much of the loss, most of all under shadow reversal and grazing light.

in Sec. VI-D) raises R@1 to 51.1%. That is still below the lunar-trained stage 1 *without* re-ranking, and only 31.9% on the hardest conditions, because the correct place is often not among the candidates at all.

C. Lunar-adapted OT aggregation

Training only the aggregation head on illumination-diverse LunarSim scenes (backbone frozen) lifts R@1 from 35.8% to 59.8% (Galatea, mean of three seeds). On the paired test queries this is a gain of 24.0 points (95% CI $[22.4, 25.7]$). The largest gains are under illumination change. Mean R@1 over the three hardest conditions (C2, C5, C6) rises from 23.4% to 47.6%. The three cluster-based heads (NetVLAD, SALAD and Galatea) clearly beat every zero-shot model. GeM pooling of the same frozen tokens does not (31.5%). Without fine-tuning the backbone, global pooling cannot separate place-specific tokens from illumination-dependent ones, while cluster-based aggregation can [10].

Galatea and SALAD are statistically indistinguishable: the paired difference in R@1 is $+0.4$ points (95% CI $[-0.5, +1.3]$). This is what Prop. 4 predicts. After intra-normalisation, the cluster-side constraint affects the descriptor only through token weights. The two heads differ sharply in how they spend mass. The balanced head sends 83.3% of all token mass to the dust-bin, and its total dust-bin mass correlates with the sky fraction of the image ($\rho = 0.63$). The learned unbalanced head ($\bar{\lambda} = 0.50$) discards only 0.4% and concentrates the remaining mass on a few clusters (coefficient of variation of cluster mass 6.4 vs. 0.18). Their retrieval performance is nevertheless the same. On this data, therefore, the assignment's marginal constraints are not the bottleneck; the training distribution is. We return to this in Sec. VI-E.

D. Global–local fusion with OT re-ranking

Table III re-ranks the top-10 candidates of Galatea (seed 0, R@1 60.8%). Two findings stand out. First, ranking by the local score alone barely helps and can hurt: no scorer improves R@1 by more than about two points, and several lose ground. The reason is that SuperPoint responses

TABLE III
STAGE-2 RE-RANKING OF THE TOP-10 GALATEA CANDIDATES (TEST, %). *Local*: SORT BY THE LOCAL SCORE ONLY. *Fused*: EQ. (12) WITH (β, κ) CHOSEN ON THE VALIDATION SCENE.

	Local		Fused	
Local scorer	R@1	R@5	R@1	R@5
none (global ranking)	–		60.8	88.0
MNN	59.2	88.2	66.0	89.5
MNN + MAGSAC++	62.4	89.2	**67.2**	90.0
OT, appearance only	57.0	89.5	65.8	90.4
OT, appearance only + MAGSAC++	61.0	88.3	63.0	88.6
LightGlue [22] (learned)	57.5	87.4	65.3	89.6
LightGlue [22] (learned) + MAGSAC++	60.0	89.0	65.7	89.8
Partial FGW (ours)	59.8	90.1	66.0	90.1
Partial FGW (ours) + MAGSAC++	63.0	88.6	63.8	88.7

TABLE IV
ABLATIONS OF THE STAGE-1 AGGREGATOR (TEST R@1/R@5, %; HARD = MEAN R@1 OVER C2, C5, C6). TOP: ASSIGNMENT RELAXATION λ (MEAN±STD OVER THE SEEDS AVAILABLE; SINGLE-SEED ROWS MARKED †). BOTTOM: GALATEA VARIANTS VS. THE SEED-0 REFERENCE.

Variant	R@1	R@5	hard
$\lambda = 0$ (NetVLAD assignment + dust-bin)	59.9±0.8	87.6±0.6	48.4
$\lambda = 0.25$ (fixed)†	60.7	88.2	48.1
λ learned (Galatea, init. 0.5) ($\bar{\lambda}$=0.50)	59.8±0.8	87.3±0.5	47.6
λ learned, 25× lr on λ, z ($\bar{\lambda}$=0.56)†	60.3	87.7	47.3
$\lambda = 0.75$ (fixed)†	59.2	87.9	48.2
$\lambda = 1$ (SALAD, balanced)	59.4±0.7	87.3±0.7	48.6
Galatea (seed 0, reference)	60.8	88.0	47.8
no dust-bin	62.1	88.0	49.3
$L = 1$ Sinkhorn iteration	59.9	87.8	46.6
$L = 10$ iterations	60.7	88.0	47.7
$m = 16$ clusters	57.7	86.2	45.9
$m = 32$ clusters	61.1	88.2	47.9
single-illumination training	47.3	72.2	27.6

on regolith are dominated by shadow edges, which move with the Sun. Second, fusing local and global evidence with Eq. (12) helps every scorer. With partial FGW, R@1 rises to 66.0%, a gain of 5.2 points (95% CI $[4.1, 6.3]$), and on the hardest conditions from 47.8% to 52.2%, the best of all scorers on that subset. Differences between scorers in the fused setting are small and within the confidence intervals: FGW vs. LightGlue $+0.7$ $[-0.6, +2.0]$, and FGW vs. appearance-only OT $+0.2$ $[-0.8, +1.1]$. The training-free structural matcher thus performs as well as a learned terrestrial matcher. MAGSAC++ helps MNN and LightGlue but not the OT matchers, whose dust-bins already reject outliers. The effect is robust across stage-1 seeds. Repeating the protocol on all three Galatea seeds (re-tuning β, κ on validation each time), fusion gains 6.3 ± 0.8 points R@1 with FGW, 6.9 ± 0.4 with MNN+MAGSAC++ and 5.2 ± 0.6 with plain MNN. The practical lesson is that, on the Moon, *how local and global evidence are combined matters more than which local matcher is used.*

E. Ablations

Table IV varies one factor at a time. Single-seed differences below about two points should be read against the seed standard deviation of roughly one point (Table II).

Assignment relaxation: Every value of λ gives the same accuracy within seed noise. R@1 is 59.9% at $\lambda = 0$ (three seeds), 59.8% with learned λ, and 59.4% at $\lambda = 1$. When λ is learned, it barely leaves its initialisation ($\bar{\lambda} = 0.50$). Even with a $25\times$ larger learning rate on λ and z it only reaches 0.56, at unchanged accuracy (60.3%). Both observations follow from Prop. 4. Once descriptors are intra-normalised, λ acts only through the token weights w_i, a second-order effect that yields a weak training signal. The practical implication is that the cheaper $\lambda = 0$ assignment (a single softmax, no Sinkhorn loop) is as good as balanced OT for this task.

Dust-bin, iterations and clusters: Removing the dust-bin does not hurt here (62.1% vs. 60.8%, single seed). In LunarSim the uninformative sky is near-black, so its tokens are already near-constant and cheap to ignore. This may not hold for real imagery with stray light or lens flare. Changing the number of Sinkhorn iterations ($L = 1$: 59.9%, $L = 10$: 60.7%) moves R@1 by less than seed noise, in line with the fast contraction of Prop. 2. Halving the clusters to $m = 32$ (4352-D) costs nothing (61.1%), which is attractive for bandwidth-limited rovers. With $m = 16$ (2304-D), R@1 drops to 57.7%.

Illumination-diverse training: By far the largest effect comes from the training data. We re-rendered the five training scenes with the *same* poses but a single Sun direction (that of the database). Galatea then reaches only 47.3% R@1, 13.6 points below illumination-diverse training, and SALAD drops the same way (46.9%). The single-illumination model is slightly *better* when the query light matches its training light (C0: 85% vs. 80%), but on the three hardest conditions it falls to 27.6%, against 47.8%. For lunar VPR, an illumination sweep in the training data, as provided by renderers such as LunarSim or DEM-based pipelines, matters far more than the choice of aggregation layer.

F. Run time

On the same 2-core CPU, the frozen ViT-B/14 takes 228 ms per image and SuperPoint 140 ms. The heads add only 1.8–3.0 ms (Galatea: 3.0 ms), so the λ-scaled Sinkhorn loop is negligible. Per candidate pair (200 keypoints), appearance-only OT takes 8.3 ms, LightGlue 19 ms and partial FGW 43 ms. FGW re-ranking of $K = 10$ candidates thus costs about 427 ms per query, so MNN+MAGSAC++ is the cheaper choice at similar accuracy.

G. Limitations

All quantitative results come from simulation. LunarSim reproduces the statistical character of lunar relief and the dominant photometric effects, but it omits several things: resolved near-field rocks below the 0.3 m DEM spacing (Fig. 2), multiple scattering beyond a constant ambient term, rover self-shadowing and dust, lens flare, exposure bracketing, and the full range of geological units. Absolute recall values should therefore not be read as predictions of on-Moon performance. The *relative* comparisons are made under identical conditions and with paired queries. Because the backbone is frozen and no method sees test scenes, we expect these comparisons to carry over better than the absolute values, but this must be

confirmed on real imagery. The backbone was frozen because of compute constraints, and joint fine-tuning may change the ranking between heads. The FGW structural term uses image-plane distances, which are only approximately preserved under large viewpoint changes. Lifting database keypoints to 3-D with DEM depth is a natural extension.

VII. Conclusion

Galatea treats both stages of lunar place recognition as optimal-transport problems. For aggregation, a semi-relaxed unbalanced formulation unifies the NetVLAD and SALAD assignments under one parameter. It comes with an explicit convergence rate, a closed-form view of the dust-bin, and an invariance result showing that intra-normalised descriptors see the plan only through token weights. For re-ranking, partial fused Gromov–Wasserstein matching adds the relative layout of keypoints to their appearance, and a saturating fusion merges local and global evidence. On the LunarSim-VPR benchmark, three findings are useful for practitioners: (i) terrestrial models, including foundation-model ones, lose more than half their recall under large Sun changes; (ii) illumination-diverse training data is worth far more than the choice of aggregation layer, and, as predicted by the theory, balanced and unbalanced assignments perform alike; (iii) fusing local with global evidence adds five to seven points of Recall@1 regardless of the local matcher, whereas re-ranking by local evidence alone helps little. The next steps are evaluation on DEM-registered real imagery, joint fine-tuning of the backbone, and 3-D structural terms that use the database's DEM depth.

References

[1] M. Maimone, Y. Cheng, and L. Matthies, "Two years of visual odometry on the Mars exploration rovers," *Journal of Field Robotics*, vol. 24, no. 3, pp. 169–186, 2007.

[2] A. E. Johnson and J. F. Montgomery, "Overview of terrain relative navigation approaches for precise lunar landing," in *Proc. IEEE Aerospace Conference*, 2008.

[3] M. Ceresoli, S. Silvestrini, and M. Lavagna, "Optical image generation through digital terrain models for autonomous lunar navigation," *Aerospace*, vol. 12, no. 2, p. 92, 2025.

[4] S. Lowry, N. Sünderhauf, P. Newman, J. J. Leonard, D. Cox, P. Corke, and M. J. Milford, "Visual place recognition: A survey," *IEEE Transactions on Robotics*, vol. 32, no. 1, pp. 1–19, 2016.

[5] S. Schubert, P. Neubert, S. Garg, M. Milford, and T. Fischer, "Visual place recognition: A tutorial," *arXiv preprint arXiv:2303.03281*, 2023.

[6] R. Arandjelović, P. Gronat, A. Torii, T. Pajdla, and J. Sivic, "NetVLAD: CNN architecture for weakly supervised place recognition," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 5297–5307.

[7] F. Radenović, G. Tolias, and O. Chum, "Fine-tuning CNN image retrieval with no human annotation," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 41, no. 7, pp. 1655–1668, 2019.

[8] G. Berton, C. Masone, and B. Caputo, "Rethinking visual geo-localization for large-scale applications," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2022.

[9] G. Berton, G. Trivigno, B. Caputo, and C. Masone, "EigenPlaces: Training viewpoint robust models for visual place recognition," in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2023.

[10] S. Izquierdo and J. Civera, "Optimal transport aggregation for visual place recognition," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 17 658–17 668.

[11] S. Hausler, S. Garg, M. Xu, M. Milford, and T. Fischer, "Patch-NetVLAD: Multi-scale fusion of locally-global descriptors for place recognition," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2021, pp. 14 141–14 152.

[12] G. Neukum, B. A. Ivanov, and W. K. Hartmann, "Cratering records in the inner solar system in relation to the lunar reference system," *Space Science Reviews*, vol. 96, pp. 55–86, 2001.

[13] H. J. Melosh, *Impact Cratering: A Geologic Process*. Oxford University Press, 1989.

[14] A. Torii, J. Sivic, T. Pajdla, and M. Okutomi, "Visual place recognition with repetitive structures," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2013, pp. 883–890.

[15] A. S. McEwen, "Photometric functions for photoclinometry and other applications," *Icarus*, vol. 92, no. 2, pp. 298–311, 1991.

[16] B. Hapke, *Theory of Reflectance and Emittance Spectroscopy*, 2nd ed. Cambridge University Press, 2012.

[17] H. Jégou, M. Douze, C. Schmid, and P. Pérez, "Aggregating local descriptors into a compact image representation," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2010, pp. 3304–3311.

[18] A. Ali-bey, B. Chaib-draa, and P. Giguère, "MixVPR: Feature mixing for visual place recognition," in *Proc. IEEE/CVF Winter Conf. Applications of Computer Vision (WACV)*, 2023, pp. 2998–3007.

[19] N. Keetha, A. Mishra, J. Karhade, K. M. Jatavallabhula, S. Scherer, M. Krishna, and S. Garg, "AnyLoc: Towards universal visual place recognition," *IEEE Robotics and Automation Letters*, vol. 9, no. 2, pp. 1286–1293, 2024.

[20] M. Oquab, T. Darcet, T. Moutakanni, H. Vo, M. Szafraniec, V. Khalidov *et al.*, "DINOv2: Learning robust visual features without supervision," *Transactions on Machine Learning Research*, 2024.

[21] P.-E. Sarlin, D. DeTone, T. Malisiewicz, and A. Rabinovich, "SuperGlue: Learning feature matching with graph neural networks," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 4938–4947.

[22] P. Lindenberger, P.-E. Sarlin, and M. Pollefeys, "LightGlue: Local feature matching at light speed," in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2023, pp. 17 627–17 638.

[23] D. DeTone, T. Malisiewicz, and A. Rabinovich, "SuperPoint: Self-supervised interest point detection and description," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2018, pp. 224–236.

[24] F. Mémoli, "Gromov–Wasserstein distances and the metric approach to object matching," *Foundations of Computational Mathematics*, vol. 11, no. 4, pp. 417–487, 2011.

[25] G. Peyré, M. Cuturi, and J. Solomon, "Gromov-Wasserstein averaging of kernel and distance matrices," in *Proc. Int. Conf. Machine Learning (ICML)*, 2016, pp. 2664–2672.

[26] T. Vayer, L. Chapel, R. Flamary, R. Tavenard, and N. Courty, "Optimal transport for structured data with application on graphs," in *Proc. Int. Conf. Machine Learning (ICML)*, 2019, pp. 6275–6284.

[27] L. Chapel, M. Z. Alaya, and G. Gasso, "Partial optimal transport with applications on positive-unlabeled learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[28] M. Cuturi, "Sinkhorn distances: Lightspeed computation of optimal transport," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2013, pp. 2292–2300.

[29] G. Peyré and M. Cuturi, "Computational optimal transport," *Foundations and Trends in Machine Learning*, vol. 11, no. 5–6, pp. 355–607, 2019.

[30] L. Chizat, G. Peyré, B. Schmitzer, and F.-X. Vialard, "Scaling algorithms for unbalanced optimal transport problems," *Mathematics of Computation*, vol. 87, no. 314, pp. 2563–2609, 2018.

[31] Y. Rubner, C. Tomasi, and L. J. Guibas, "The earth mover's distance as a metric for image retrieval," *International Journal of Computer Vision*, vol. 40, no. 2, pp. 99–121, 2000.

[32] Y. Zhang, Z. Kang, and Z. Cao, "An image retrieval method for lunar complex craters integrating visual and depth features," *Electronics*, vol. 13, no. 7, p. 1262, 2024.

[33] X. Wang, X. Han, W. Huang, D. Dong, and M. R. Scott, "Multi-similarity loss with general pair weighting for deep metric learning," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 5022–5030.

[34] D. Barath, J. Noskova, M. Ivashechkin, and J. Matas, "MAGSAC++, a fast, reliable and accurate robust estimator," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 1304–1312.

[35] NASA, "Apollo 11 image library," https://www.nasa.gov/history/alsj/a11/images11.html, 2020.

[36] A. Ali-bey, B. Chaib-draa, and P. Giguère, "GSV-Cities: Toward appropriate supervised visual place recognition," *Neurocomputing*, vol. 513, pp. 194–203, 2022.